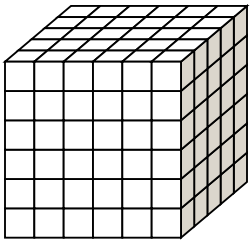
Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 93178

Part A: Each question is worth 3 points

1. Chris built this brick (6 cubes long, 5 cubes deep and 6 cubes tall) from red and blue cubes of the same size. Every cube you can see from the outside (also from below and behind) is red. All the cubes inside are blue.



How many red cubes did Chris use?

A. 48 ***B.*** 124 ***C.*** 132 ***D.*** 180 ***E.*** 192

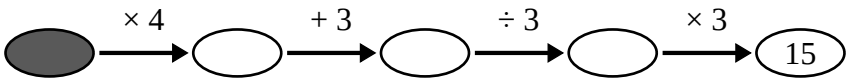
2. The odometer of a bike shows 31713 km. This number reads the same forwards and backwards. After how many more kilometres will this happen again?

A. 10 ***B.*** 11 ***C.*** 90 ***D.*** 100 ***E.*** 200

3. Sara has the cards 2, 8 and 9. Sara must use every card exactly once, with +, –, ×, ÷ and brackets. Which of these numbers can Sara get?

A. 20 ***B.*** 22 ***C.*** 26 ***D.*** 29 ***E.*** 32

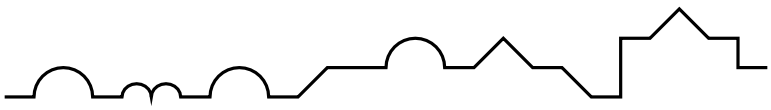
4. Leo wrote a favourite number in the dark cloud and did the calculations shown by the arrows, one after the other. The result was 15.



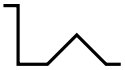
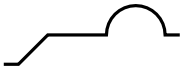
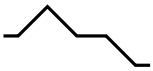
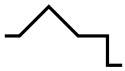
What is Leo's favourite number?

A. 3 ***B.*** 4 ***C.*** 5 ***D.*** 15 ***E.*** 63

5. This is the outline of a town seen from the river at sunset.



Which piece **cannot** be part of the outline?

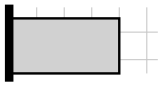
A.  ***B.***  ***C.*** ***D.***  ***E.*** 

Part B: Each question is worth 4 points

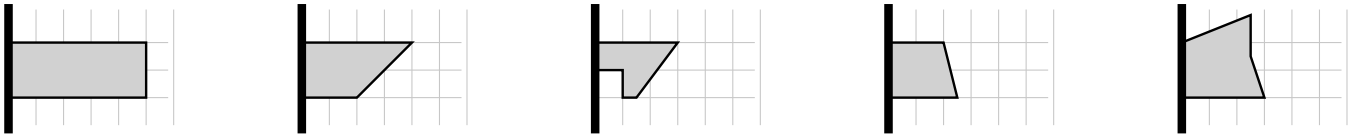
6. What is the smallest number whose digits add up to 24?

A. 617 ***B.*** 625 ***C.*** 629 ***D.*** 699 ***E.*** 700

7. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Mia draws the shadow at five different moments. Flat, the flag is 4 squares wide and 2 squares tall (see the picture). Some of the drawings are wrong.



the flag, flat



A B C D E

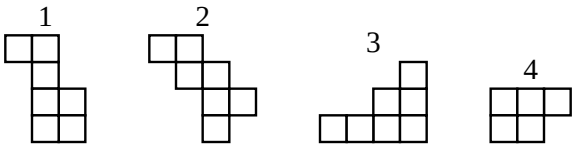
How many of the five drawings can be right?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

8. The rooms in a hotel are numbered from 1 to 110.
How many times is the digit 3 written?

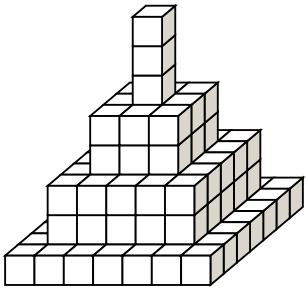
A. 10 ***B.*** 20 ***C.*** 21 ***D.*** 22 ***E.*** 26

9. Which two of these pieces fit together to make a rectangle? Pieces may be turned and flipped over.



A. 2 and 3 ***B.*** 1 and 4 ***C.*** 3 and 4 ***D.*** 1 and 3 ***E.*** 1 and 2

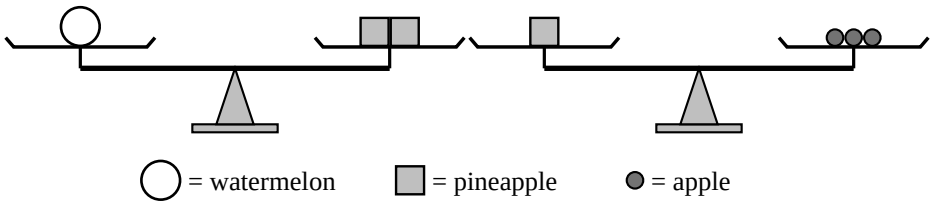
10. Robert built this pyramid out of identical cubes. Every layer is a full square, with no holes hidden inside.



How many cubes are in the pyramid?

A. 120 ***B.*** 127 ***C.*** 133 ***D.*** 136 ***E.*** 141

11. The two balances are level. One watermelon weighs as much as 2 pineapples, and one pineapple weighs as much as 3 apples.



How many apples weigh as much as 3 watermelons?

A. 5 ***B.*** 6 ***C.*** 7 ***D.*** 9 ***E.*** 18

Part C: Each question is worth 5 points

12. A bar code is made of 22 black and white bars taking turns. It starts with a black bar and ends with a **white** bar. Each black bar is wide or narrow. There are 7 more white bars than wide black bars.
How many narrow black bars are there?

A. 11 ***B.*** 10 ***C.*** 9 ***D.*** 8 ***E.*** 7

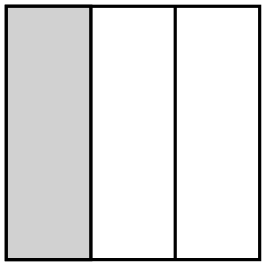
13. Five children, David, Clara, Hugo, Felix and Ben, sit in a row of 5 chairs at the cinema. David and Clara are best friends and want to sit next to each other.
In how many different ways can the children sit?

A. 72 ***B.*** 48 ***C.*** 24 ***D.*** 8 ***E.*** 4

14. Sara has letter cards and wants to show which letters are used in the words TEACHER and TURTLE. Sara puts down one card for each different letter used in either word.
How many cards does Sara put down?

A. 7 ***B.*** 8 ***C.*** 10 ***D.*** 11 ***E.*** 13

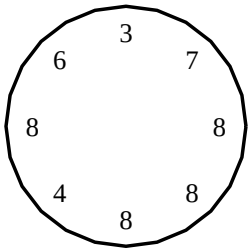
15. Mia cuts a square chocolate bar into 3 equal strips as shown. The perimeter of each strip is 80 cm.



What is the perimeter of the square?

A. 240 cm ***B.*** 160 cm ***C.*** 134 cm ***D.*** 121 cm ***E.*** 120 cm

16. Numbers are written around a round plate as shown. Tom wants to break the plate with one straight line so that the two pieces have the same sum of numbers.



In how many different ways can he do it?

A. 0 ***B.*** 1 ***C.*** 3 ***D.*** 4 ***E.*** 5

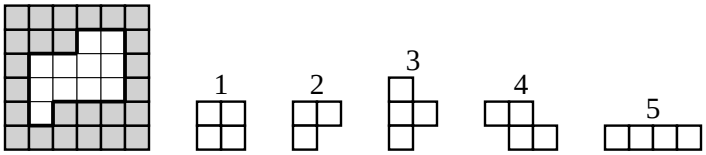
17. Eva's new rug is 1 m 9 cm long. Four of these lengths are the same as that.
Which one is *not* the same?

A. 109 cm ***B.*** 1 m 90 mm ***C.*** 10 dm 9 cm ***D.*** 190 cm ***E.*** 1090 mm

18. Sofia, Emma, Manon, Lucas and Hana were born in different months: August, October, December, June and January. Emma has a birthday in summer. Lucas has a birthday in summer. Hana was born in autumn. Lucas was born in a month that starts with J. Sofia was born in a month that starts with J.
Which statement is **not** true?

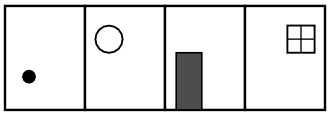
A. Hana was born in October ***B.*** Sofia was born in January ***C.*** Lucas was born in December
D. Emma was born in August ***E.*** Manon was born in December

19. Which three of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.

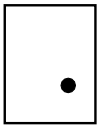
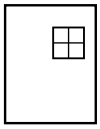
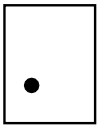
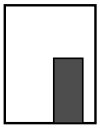
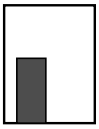


A. 1+2+3 ***B.*** 1+2+4 ***C.*** 1+4+5 ***D.*** 2+3+4 ***E.*** 2+4+5

20. The strip below is cut out and folded into the four walls of a house, with the pictures on the outside. A flat roof is put on top. You stand outside the house at a corner. The wall with the round window is on your left.



Which wall do you see on your right?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

Answer Key

1.	**C**	**2.**	**D**	**3.**	**C**	**4.**	**A**	**5.**	**A**	**6.**	**D**
7.	**D**	**8.**	**C**	**9.**	**C**	**10.**	**A**	**11.**	**E**	**12.**	**E**
13.	**B**	**14.**	**B**	**15.**	**E**	**16.**	**B**	**17.**	**D**	**18.**	**C**
19.	**D**	**20.**	**E**								

Solutions

1. C (132) — All cubes: 6 × 5 × 6 = 180. Blue inside: 4 × 3 × 4 = 48. Red: 180 – 48 = 132. [2003 Q9 · red_outside]

2. D (100) — The next such number is 31813, so 31813 – 31713 = 100. [2002 Q24 · next_palindrome]

3. C (26) — 2 × 9 + 8 = 26. None of the other numbers can be made. [2002 Q21 · which_can]

4. A (3) — Go backwards and do the opposite: 15 ÷ 3 = 5, 5 × 3 = 15, 15 – 3 = 12, 12 ÷ 4 = 3. [2007 Q5 · favourite_number]

5. A — All the other pieces can be found in the line, but this one cannot: it steps down where the line steps up. [2002 Q7 · cannot_belong]

6. D (699) — Use as few digits as possible: as many 9s as possible, with the smallest digit in front. 699: 6 + 9 + 9 = 24. [2007 Q4 · smallest_with_sum]

7. D (3) — A drawing is wrong if part of the pole edge is bare or if it reaches more than 4 squares from the pole. Wrong: A, C. So 3 can be right. [2002 Q15 · count_possible]

8. C (21) — As the last digit: 11 times; as the tens digit: 10 times. Total 21. [2003 Q16 · count_one_digit]

9. C (3 and 4) — Pieces 3 and 4 have 12 squares together and fit into a 3 × 4 rectangle. No other pair does. [2007 Q8 · two_make_rectangle]

10. A (120) — Count layer by layer, from the bottom: 7 × 7 + 5 × 5 × 2 + 3 × 3 × 2 + 1 × 1 × 3 = 120. [2002 Q20 · count_pyramid]

11. E (18) — One watermelon = 2 pineapples = 2 × 3 = 6 apples; 3 watermelons = 3 × 6 = 18 apples. [2002 Q9 · chain]

12. E (7) — Starting black and ending white, the bars come in pairs: 11 black and 11 white. Wide: 11 – 7 = 4. Narrow: 11 – 4 = 7. [2003 Q17 · ends_differ]

13. B (48) — Think of David and Clara as one block: the block and the other 3 make 4 things to order, 4 × 3 × 2 × 1 = 24 ways. Inside the block David and Clara can swap, so 2 × 24 = 48. [2002 Q22 · side_by_side]

14. B (8) — TEACHER uses 6 different letters and TURTLE uses 5. E, R and T are in both, so count them once: 6 + 5 – 3 = 8. [2007 Q3 · letter_cards]

15. E (120 cm) — A strip is the full side long and 1/3 of it wide, so its perimeter is side + side + 2 × (side ÷ 3) = 8 × (side ÷ 3). 80 ÷ 8 = 10, so the side is 3 × 10 = 30 cm and the perimeter is 120 cm. [2002 Q6 · strips_to_square]

16. B (1) — Each half must have 26. The ways: 3+7+8+8 | 8+4+8+6. [2002 Q11 · count_ways]

17. D (190 cm) — 1 m 9 cm = 109 cm = 1090 mm. But 190 cm = 190 cm. [2003 Q10 · not_equal]

18. C (Lucas was born in December) — The only possible matching: Sofia – January; Emma – August; Manon – December; Lucas – June; Hana – October. [2002 Q18 · which_not_true]

19. D (2+3+4) — The empty area has 11 squares, so the three sizes must add up to 11. Only 2 + 3 + 4 also fit together in the right shape. [2003 Q18 · three_pieces]

20. E — Folded with the pictures outside, the strip keeps its order: on your right is the next wall on the strip. Its mark stays on the same side, so it is picture E. [2003 Q12 · right_neighbour]